From Revit to Robot: BIM-Derived Digital Twins for Sim-to-Real Autonomous Building Patrol

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Abstract

Autonomous indoor service robots are normally trained only after a building is finished and accessible, which is slow, costly, and risky on a live campus. We present ***Revit to Robot****, an end-to-end pipeline that turns a building's architectural design data into a physics-accurate training ground for an autonomous mobile robot* before the physical structure is complete. *Starting from Revit/BIM models of the MSOE campus, we automatically convert architecture into simulation-ready USD digital twins in NVIDIA Isaac Sim, applying triangle-mesh colliders to static structure (walls, floors, glazing, stairs) and articulated revolute joints with calibrated hinge offsets and motor torque to make every door functional. From each twin we extract a 2D occupancy map by ray-tracing the model from 125 vantage points (*$129\,m \times 142\,m$*,* 98.4% *floor coverage), place a navigation graph with NVIDIA SWAGGER, and compute an efficient door-inspection patrol route via Dijkstra distance tables refined with greedy nearest-neighbor and 2-opt search (a traveling-salesman formulation). A differential-drive Nova Carter robot ("ROSCOE") drives this route with the ROS 2 Nav2 stack, using its onboard* 360° *LiDAR for real-time local planning and obstacle avoidance between waypoints. For environmental awareness, we fine-tune a Qwen3-VL-2B vision-language model with LoRA (*$\approx 1.6\%$ *of parameters trained) to detect the two highest-priority anomalies identified by campus public safety — open/unsecured doors and cracked glass — using only synthetic imagery generated by programmatically manipulating the digital twin's internal state, which yields free ground-truth labels across thousands of randomized lighting, material, and viewpoint conditions. The fine-tuned model reaches* 97.4% */* 97.2% *accuracy on held-out door / glass frames, retains* 90.2% *on a held-out-instance door test, and transfers to real-world imagery of an already-built campus building at* 93.0% *accuracy. The entire system is repeatable for any building with BIM data; we apply it to a six-building digital campus, consuming over 3,500 GPU jobs and 2,000 GPU-hours on MSOE's Rosie supercomputer. Our results show that BIM data — typically treated as static documentation — can serve as a scalable, automatically labeled training substrate for sim-to-real robot autonomy.*

1. Introduction

Simulation-based training has become the dominant approach for developing robot navigation systems. It offers safe, scalable, and repeatable experimentation without the cost or risk of real-world trials. However, most robots are trained in generic virtual scenes that do not match the specific geometry of their actual deployment sites. Building these high-fidelity environments by hand is slow, expensive, and requires specialized expertise. This bottleneck limits simulation-based robot training to well-resourced teams and constrains most deployed systems to generic environments that do not reflect where they will actually operate.

Building Information Modeling (BIM) offers a path forward that has gone largely unexploited in the robotics community. BIM tools such as Autodesk Revit are already used by architects and facility managers to produce precise, semantically rich 3D models of buildings as part of standard construction and maintenance workflows. These models contain exactly the geometric and spatial information needed to construct a simulation environment, yet this data is rarely connected to robot training.

This paper presents a framework to bridge that gap. Our pipeline converts Revit files into physics-accurate digital twins in NVIDIA Isaac Sim without manual scene authoring. From each twin we (i) extract a 2D occupancy map and a navigation graph for deterministic patrol-route planning, (ii) drive a differential-drive robot along those routes with the ROS 2 Nav2 stack for real-time, LiDAR-based obsta-

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cle avoidance, and (iii) fine-tune a vision-language model (VLM) for anomaly detection using only synthetic imagery whose ground-truth labels come for free from the twin's internal physics and material state.

We demonstrate the framework on autonomous security patrol — a domain where the need for site-specific training is acute and the connection to real facility data is natural. The system is developed in coordination with MSOE's Public Safety Office and is anchored to the Robert D. Kern Engineering Innovation Center, a four-story facility currently under construction. Because the building does not yet physically exist, it cannot be entered, photographed, or used for data collection — making a BIM-derived digital twin not merely convenient but the only way to train for it ahead of opening.

An earlier version of this system trained a reinforcement-learning (RL) policy in Isaac Lab for local navigation. We found that a classical, well-engineered navigation stack (ROS 2 Nav2) reached reliable obstacle avoidance and route following far faster, with no reward engineering and a smaller sim-to-real gap for locomotion, so Nav2 is the navigation method used throughout this paper; the RL effort is retained only as an archived exploration.

The contributions of this paper are:

1. **An end-to-end pipeline from architectural BIM data to a simulation-ready digital twin** in NVIDIA Isaac Sim, requiring no manual scene authoring, and scaled to a six-building digital campus.
2. **An automatic navigation-graph and patrol-route pipeline** that derives a 2D occupancy map by ray-tracing the twin, places a navigation graph with NVIDIA SWAGGER, and computes efficient door-inspection routes as a traveling-salesman problem (Dijkstra distance tables + greedy nearest-neighbor + 2-opt).
3. **A deployment of the ROS 2 Nav2 stack inside the BIM-derived twin** that drives a Nova Carter robot through planned routes using onboard LiDAR for local planning and obstacle avoidance, with documented parameter choices that make off-the-shelf Nav2 work in long architectural corridors.
4. **A label-free VLM fine-tuning methodology for sim-to-real anomaly detection** that adapts a Qwen3-VL-2B model with LoRA and vision-projector co-training to detect two public-safety anomalies — open/unsecured doors and cracked glass — from synthetic renders alone, and that transfers to real-world building imagery at over 90% accuracy.

2. Related Work

Simulation-based navigation and Isaac Sim. Simulation-based training is a cornerstone of mobile-robot development because it enables policy learning in safe, repeatable environments before physical deployment, with extensive work on obstacle avoidance and goal-directed indoor navigation [11]. NVIDIA Isaac Sim has become a leading platform for high-fidelity robot simulation, with Isaac Lab extending it for massively parallel learning [8], and Salimpour *et al.* [10] demonstrated end-to-end sim-to-real transfer of navigation policies on physical ROS 2 robots using this stack. A limitation shared by these and most simulation-based approaches is that the training environments are generic, hand-authored scenes with no relationship to the actual deployment site. Our framework replaces the hand-authored environment with one automatically derived from the target building's own architectural data.

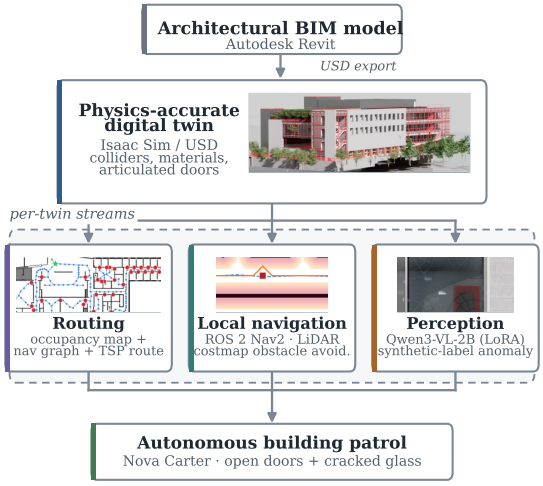


Figure 1. Overview of the integrated BIM-to-Robot pipeline. A single Autodesk Revit model becomes a physics-accurate Isaac Sim digital twin that feeds three streams — navigation-graph routing, ROS 2 Nav2 local navigation, and Qwen3-VL anomaly perception — converging on an autonomous Nova Carter patrol.

Classical navigation (Nav2). Rather than learn a local-navigation policy from scratch, we drive the robot with the ROS 2 Nav2 stack [6], the de facto standard for autonomous ground-robot navigation. Nav2 composes a global planner, a local trajectory controller, probabilistic localization (AMCL), and a behavior tree for recovery, over layered costmaps fused from LiDAR. Its components are mature and well-studied, which lets us focus the research contribution on the BIM-to-twin and perception pipelines rather than on re-deriving local navigation.

BIM for robot navigation. BIM has been explored as a source of spatial knowledge for robot navigation, with prior work showing that BIM/IFC data can generate navigation maps, occupancy grids, and waypoint sequences for autonomous inspection [1, 3, 5, 9]. A fundamental limitation shared across these works is the extraction of simplified 2D or graph-based representations rather than preserving full

3D geometry, which precludes high-fidelity physics simulation or in-twin policy training. Our framework converts BIM geometry to the Universal Scene Description (USD) format and loads the complete 3D scene into Isaac Sim, enabling physics-accurate simulation, articulated doors, and onboard sensor rendering inside the as-designed building.

Vision-language models and parameter-efficient fine-tuning. VLMs enable semantic scene understanding without task-specific classifiers, but models pretrained on real imagery suffer a sim-to-real gap when applied to synthetic renders. Parameter-efficient methods, particularly Low-Rank Adaptation (LoRA) [4], adapt large models to new visual domains while preserving general reasoning. We apply LoRA to Qwen3-VL-2B-Instruct [12] and additionally co-train the vision projection layer, bridging Isaac Sim's synthetic visual tokens to the model's pretrained embedding space. Among small open VLMs we evaluated zero-shot (Sec. 4.4), Qwen3-VL-2B was both the strongest baseline and the smallest deployable choice, motivating it as our backbone.

3. System Architecture

The framework is organized as an integrated multi-stream pipeline (Fig. 1). The process begins by converting an Autodesk Revit BIM file into USD to create a physics-accurate digital twin in NVIDIA Isaac Sim, where physics materials, lighting, and sensor configurations are applied. This environment is the foundation for three streams: (i) occupancy-map extraction and navigation-graph route planning for deterministic patrol coverage, (ii) ROS 2 Nav2 for local planning and obstacle avoidance, and (iii) VLM adaptation for semantic anomaly detection. The streams converge into a unified system ready for deployment on the physical NVIDIA Nova Carter platform. While the framework generalizes to any BIM-equipped facility, we demonstrate it through an autonomous security patrol in coordination with MSOE's Public Safety Office.

3.1. Building the Dataset

The Milwaukee School of Engineering's new Robert D. Kern Engineering Innovation Center is under construction and slated for completion in fall 2027. The building spans four stories and over 97,000 square feet at a cost of roughly $76.5 million, with modern labs, classrooms, workshops, and collaborative spaces oriented around artificial intelligence. The design files were provided to us by MSOE from Ramlow Stein Architecture and Interior in Revit 2024 format [7]. The Revit file was 818 MB and was pared down to $\approx$ 35 MB by deleting elements unnecessary for navigation and perception; the elements retained were walls, floors, ceilings, railings, doors, and stairs. Fig. 2 shows the resulting digital twin rendered in Isaac Sim, with the building's signature red-mullioned curtain wall and brick massing preserved from the architectural model.



Figure 2. The Robert D. Kern Engineering Innovation Center as a physics-accurate digital twin in NVIDIA Isaac Sim, rendered from the BIM-derived USD. The red-mullioned curtain wall, brick massing, landscaping, and per-floor interior structure are carried over directly from the Revit model — the same geometry the robot navigates and the onboard camera perceives.

3.2. Scene Preparation

3.2.1. BIM-to-USD Conversion

A major challenge was converting the Revit 2024 model to USD while preserving the detail needed for downstream training. To optimize simulation performance, the Revit model filtered non-essential systems (HVAC, electrical, plumbing), retaining only structural components such as walls, floors, and doors. This reduced file size from 818 MB to $\approx$ 35 MB and cut the number of elements Isaac Sim must render. We exported with the NVIDIA Omniverse Connector with the *"Family Data Instancing"* setting enabled — critical for preserving individual building components as discrete objects (rather than a single fused mesh), which maintains the semantic properties and structural hierarchy needed for high-fidelity training and automatic per-element physics classification.

3.2.2. Isaac Sim Scene and Physics

The building USD is imported into Isaac Sim and organized as a reusable world reference, so the architectural model functions as a simulation-ready environment. Physics is then applied on top of NVIDIA PhysX, providing gravity, rigid-body simulation, collision handling, and joint-based motion. Static elements (walls, floors, glazing, stairs) are assigned **triangle-mesh colliders** so the environment acts as a navigable physical space, while doors are configured as **articulated rigid bodies with revolute joints and drive controllers** for realistic opening behavior. Materials are grouped by surface type so major components receive appropriate physical properties, and lighting and rendering are configured through Isaac Sim's standard world-

initialization workflow.

Doors required particular care. An initial approach that guessed the hinge location produced doors that clipped into walls. The working solution places the hinge at the vertical edge of the door with a slight offset to avoid colliding with the door's bounding box, and applies a drive motor whose minimum torque is set from the door's moment of inertia and width so the leaf opens cleanly on command. The full physics process — element categorization, collider and material assignment, and door articulation — is implemented as a **replicable, automated pipeline** that runs on any Revit-exported USD, which is what makes the six-building digital campus feasible (Sec. 4).

3.3. Occupancy Map, Navigation Graph, and Route Planning

Patrol navigation is built in three stages: an occupancy map is generated by ray-tracing the twin, a navigation graph is placed on it with NVIDIA SWAGGER, and patrol routes are computed over the graph.

Occupancy mapping. For each floor, the USD scene is loaded into Isaac Sim and the `isaacsim.asset.gen.omap` extension performs a raycast-based scan from a grid of origin points. Each cell is labeled free, occupied, or unknown. Door primitives are excluded from the collision layer during scanning so rays can pass through doorways and reach inside rooms. The scanner runs multiple passes, seeding additional origin points inside remaining unknown regions and stopping automatically once less than $\sim$ 0.5% of the floor grid is unknown. Across the Innovation Center this yields **125 scan origins** over a $129\,\mathrm{m} \times 142\,\mathrm{m}$ footprint with $\mathbf{98.4}\%$ floor coverage, leaving under 2% unknown per floor (0.6% on floor 4). Maps are produced at 0.05 m/pixel and exported as a grayscale PNG (black = wall, white = free, grey = unknown), a NumPy array, and a metadata file. The PNG is written in the ROS `map_server` convention (row 0 = world $y_{\max}$; origin is the world pose of the bottom-left pixel) so it can be consumed directly by Nav2 without remapping.

Navigation graph (NVIDIA SWAGGER). The binary occupancy map is passed to NVIDIA SWAGGER, which places navigation nodes through a multi-stage pipeline: (i) *skeletonization* (a medial-axis transform) reduces corridors to one-pixel-wide center lines; (ii) *skeleton sampling* places nodes at fixed endpoints and junctions plus regular intervals along branches; (iii) *free-space sampling* adds nodes inside large open areas such as atriums and lecture halls; (iv) *triangulation* introduces more direct edges between mutually visible nodes; and (v) *pruning* removes redundant nodes/edges and discards disconnected subgraphs shorter than a minimum length. Because skeletonization is sensitive to visual noise, a clean binary occupancy map (not a raw architectural drawing with text, door arcs, and dashed lines) is required to avoid tangled graphs. The resulting per-floor graphs range from 219 nodes / 226 edges (floor 4) to 586 nodes / 589 edges (floor 3). Node positions are stored in world coordinates with edges pre-weighted by Euclidean distance, so the graph is usable in simulation without further coordinate conversion.

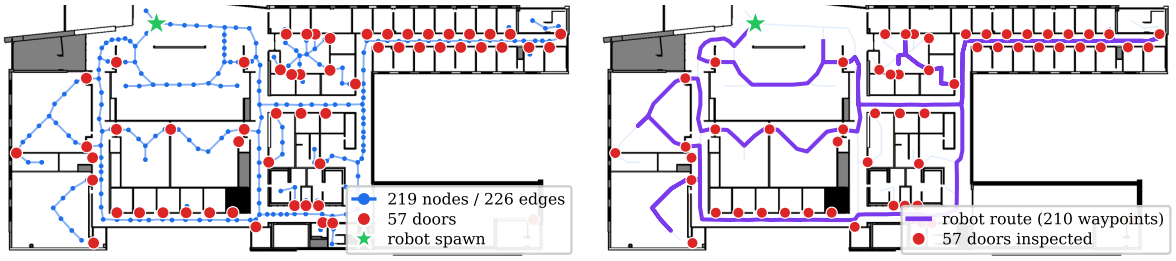
Door labeling and route planning. Using each door's position, the nearest graph node is labeled a *door node* (e.g., 57 doors collapsing to 47 unique door nodes on floor 4). A patrol that inspects every door is then a traveling-salesman problem (TSP) over the door nodes. We first run **Dijkstra's algorithm** [2] from each node to build a complete all-pairs distance table that reflects true travel distance through corridors. We generate an initial tour with a **greedy nearest-neighbor** heuristic, improve it with **2-opt** local search (which uncrosses self-intersecting routes by removing two edges and reconnecting), and repeat from multiple starting nodes (**random restart**, 10 trials), keeping the best tour. TSP is NP-hard, so this yields an efficient — not provably optimal — route. Finally, each door-to-door leg of the chosen tour is expanded into a full corridor path via Dijkstra shortest path, and the entire expanded waypoint sequence is handed to the navigation stack as a single goal (Sec. 3.4).

3.4. Local Navigation with ROS 2 Nav2

Between waypoints, the robot drives itself with the **ROS 2 Nav2** stack (ROS 2 Jazzy). Isaac Sim and Nav2 communicate through Isaac Sim's built-in ROS 2 bridge: the simulator publishes sensor streams and odometry, Nav2 publishes velocity commands, and the simulated Nova Carter executes them. The TSP-expanded route is sent as one `NavigateThroughPoses` action goal, so the robot drives the full patrol continuously rather than stopping at each waypoint.

Robot platform. The NVIDIA Nova Carter is a differential-drive mobile robot ($0.722\,\mathrm{m} \times 0.500\,\mathrm{m} \times 0.556\,\mathrm{m}$, 51.2 kg) on the Segway RMP Lite 220 chassis (wheel radius 0.14 m, wheel base 0.4 m). Its sensor suite includes a Hesai 3D LiDAR, front and rear 2D scanning LiDARs, four stereo cameras, and an IMU, with onboard compute from an NVIDIA Jetson AGX Orin. We use it because it has a validated USD model in the NVIDIA asset library and a native ROS 2 driver, so the same platform and command interface are used in simulation and (planned) physical deployment, minimizing the sim-to-real gap for locomotion. For perception we additionally mount a forward-facing first-person RGB camera (640×480, 20 fps, 5 mm focal length) that always points in the direction of travel; this is the input to the VLM (Sec. 3.5).

Sensing and localization. The Hesai 3D LiDAR point cloud is projected to a 2D `LaserScan` (`/scan`) via



(a) Navigation graph + doors (b) Door-inspection route (TSP + Dijkstra)

Figure 3. Occupancy map, navigation graph, and optimized door-inspection route on Innovation Center floor 4. (a) the SWAGGER navigation graph (219 nodes / 226 edges) with the 57 doors marked at their true world positions (each door is assigned the nearest graph node, giving 47 unique door nodes for the tour); (b) the door-inspection tour (greedy nearest-neighbor + 2-opt) expanded by Dijkstra shortest paths into the 210-waypoint route the robot actually drives through the corridors.

Parameter	Upstream	Ours	Effect
`collision_monitor min_points`	6×10^9	**4**	Re-enables the emergency stop
DWB `Base Obstacle.scale`	0.02	**15.0**	Robot swerves around obstacles
obstacle / raytrace range	10.0 m	**15.0 m**	Plans global detours early
`cost_travel_multiplier`	2.0	**4.0**	Prefers wall-clearing cells

Table 1. Deliberate Nav2 parameter changes relative to the upstream Nova Carter defaults, with their effect in long architectural corridors.



(a) Follow camera: Nova Carter at a patrol waypoint (b) Onboard first-person camera (VLM input)

Figure 4. The deployed stack in action: real frames from a floor-4 patrol run in which ROS 2 Nav2 autonomously drives the Nova Carter robot between inspection waypoints in the BIM-derived twin. (a) Third-person follow camera with the robot stopped at a glass-door waypoint; (b) its forward-facing first-person camera—the input to the VLM (Sec. 3.5)—viewing the closed glass door it has reached. Both are rendered sensor streams from the same run, not illustrations.

`pointcloud_to_laserscan` and feeds the global obstacle costmap; the front/rear 2D LiDARs feed the local costmap. Localization uses **AMCL** (Adaptive Monte Carlo Localization) against the static map produced in Sec. 3.3, with a differential motion model. The global planner is **SmacPlanner2D** (A* with cost-weighted edge selection) and the local controller is the **Dynamic Window Approach** (DWB); recovery and replanning are handled by the standard Nav2 behavior tree. Costmaps are layered: a global static + obstacle + inflation costmap over the whole floor, and a rolling 6 m $\times$ 6 m local costmap fusing the 2D scans, a 3D voxel layer, and inflation.

Parameter choices that matter. Off-the-shelf Nav2 parameters did not work in long architectural corridors; the changes in Tab. 1 were needed and are reported so others can reproduce them. Maximum commanded velocity is 1.6 m/s linear and 1.1 rad/s angular. Goal tolerance is 0.25 m / 0.25 rad with a stateful goal checker so a multi-pose tour tolerates small drift.

3.5. Vision-Language Perception

To give the robot semantic awareness, we fine-tune a vision-language model to classify anomalies in the onboard camera stream. We reached out to MSOE Public Safety for the anomalies they look for during walkthroughs and targeted the **top two**: doors left **open/unsecured** and **cracked glass**. The model is asked a single four-way question — *door open, door closed, glass cracked, or glass intact* — so one model covers both anomaly types while still describing the rest of the scene for free.

Why a VLM. Unlike a CNN classifier, a VLM already knows what a hallway, student, or fire extinguisher is, so fine-tuning only the anomaly behavior leaves its general world knowledge intact, and it can explain a flag in natural language to a human operator rather than emit a bare probability.

Synthetic data from the twin. Because the building is not yet built, we cannot photograph it; and even for built spaces

one cannot stage break-ins, floods, or fires on demand. Instead we manipulate the twin's internal state to *program* each anomaly and read the ground-truth label directly from the simulator — a free, exact label for every frame. Doors are articulated revolute joints, so the ground-truth door angle is queried from the joint and thresholded ($\leq 5^\circ$ closed, $> 5^\circ$ open). Cracked glass is rendered by swapping the material on a random subset of glass panels to a cracked variant each frame; Replicator emits the corresponding semantic-segmentation labels.

The **cracked-glass pipeline** is visual-only (no geometry fracture). Of the $\sim$ 1,013 glass panels in the Innovation Center USD, the standard double-pane, curtain-wall, and glazed system panels are crackable (shadowbox and laminated panels are excluded). Each cracked panel is rendered with an OmniGlass shader whose `normal_map_texture` input is wired to a crack normal map, swapped per frame through direct material binding so the binding and the semantic label stay consistent. Three crack styles provide variety: a physics-grounded *tempered* style (Voronoi tessellation matching real tempered-glass fracture), an *impact* style (spider-web cracks generated by Stable Diffusion XL line-art passed through a Sobel filter into a normal map), and an *edge* style (a branching random walk from a panel edge), all illustrated on the same panel in Fig. 5. Domain randomization spans lighting presets, the fraction of panels cracked per frame, the tiling/impact-point toggle, and camera pose.

The **door pipeline** holds a door at a fixed angle per episode and captures frames from a multi-camera arc; per-episode angle randomization produces a smooth spread of openness. Critically, the dataset spans **multiple doors and door families** — 8 training doors across Revit families (CW_F, CW_FG, Type_F, Type_N, Type_FG), plus 2 held-out doors (an unseen instance and an unseen swing direction) — under six lighting presets and varied viewpoint/height. This multi-door, randomized design is a direct response to a generalization failure in an earlier single-door dataset (Sec. 4.4).

Model and fine-tuning. The backbone is **Qwen3-VL-2B-Instruct**. We apply **LoRA** (rank 16, α 32, dropout 0.05) to the language backbone's `q_proj` and `v_proj`, and additionally **co-train the full vision projector** (the `visual.merger` patch-merger) via PEFT's `modules_to_save`. The projector is the bridge between visual patch tokens and the LLM's embedding space; the base projector was optimized for real photographs, and without adapting it the LLM receives no meaningful information about synthetic door/glass state — no amount of LoRA training compensates. In total $\approx \mathbf{1.62}\%$ of parameters are trained ($\approx$ 2.2M LoRA + $\approx$ 34M projector). Training runs on a Tesla V100 (Rosie `dgx` partition). Two engineering pitfalls were essential to get right: PEFT's default save drops the trained projector weights unless they are listed in `modules_to_save` (the symptom is a model that silently reverts to the untrained baseline), and trainable projector parameters must be cast to fp32 for AMP gradient scaling.

4. Evaluation

4.1. Scene Fidelity

The converted USD scene preserves the source Revit model's structural elements, semantic metadata, and spatial organization, which lets the automated physics pipeline classify elements by parent folder rather than by manual annotation. Functional accuracy is verified through three automated strategies. *Drop testing* discovers floor levels by clustering floor-element bounding boxes by Z-coordinate, spawns rigid bodies above each surface, and samples their Z over time to distinguish objects that come to rest (solid collider) from objects that fall through (missing collider). *Robot-in-the-loop validation* drives a robot in a patrol pattern on each surface while monitoring its Z, exposing surface gaps during active motion. *Automated visual validation* records an orbital 360° sweep for manual inspection of overall form and physics behavior across levels.

4.2. Navigation Graph Coverage

Occupancy maps were produced for all four above-ground floors. Unknown cells ranged from 0.6% (floor 4) to under 2% (floor 1), with 98.4% overall floor coverage at 0.05 m/pixel. We validated floor 4 by overlaying the computed world coordinates onto the occupancy image and confirming in NVIDIA Omniverse that every navigation node falls in free space, not inside walls or obstacles. Route correctness was demonstrated with Dijkstra's shortest path traversing the full layout, and full-floor reachability was confirmed by the TSP solution (greedy nearest-neighbor + 2-opt + random restart) visiting every door node.

4.3. Navigation and Route Execution

Driving the SWAGGER route with Nav2 inside the twin, the Nova Carter follows the planned door-inspection tour and avoids obstacles in real time using its LiDAR costmaps. We exercise the stack at three route scales — single-wing (easy), cross-wing (medium), and full-building (hard) — and with adversarial obstacles spawned in corridors at known world coordinates. With the tuned parameters of Sec. 3.4 the robot completes routes and recovers from blocked corridors via the Nav2 behavior tree; with the upstream defaults it either disables its emergency stop or drives through obstacles, motivating the changes we report.

A full quantitative characterization — per-scale route completion rate, mean path length relative to the graph-optimal tour, number of recovery behaviors triggered, and obstacle-avoidance success rate — is the focus of an ongo-

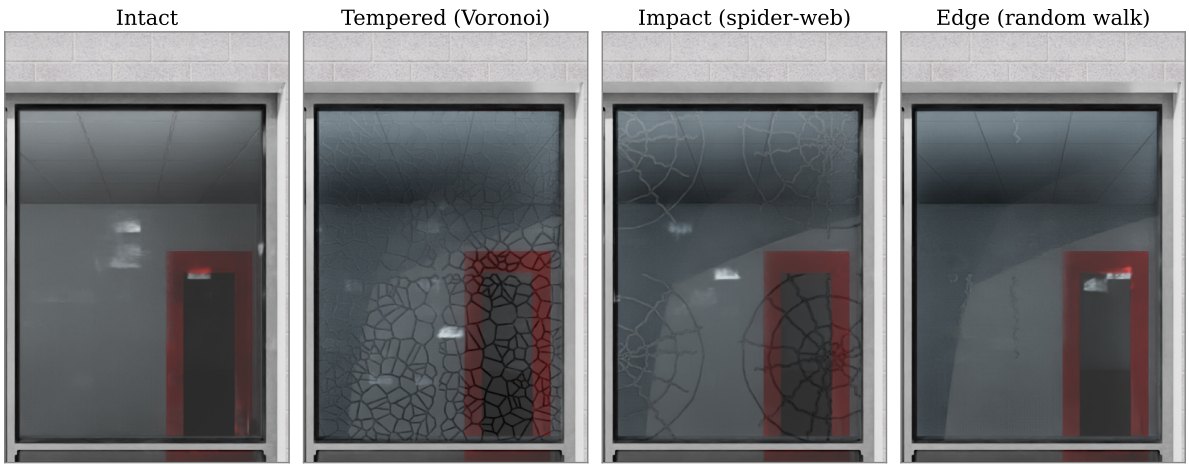


Figure 5. Synthetic anomaly data rendered from the digital twin's onboard robot camera. The same curtain-wall panel is shown intact and with the three crack styles (tempered Voronoi, impact spider-web, edge random-walk). Because the anomaly is programmed into the twin, every frame carries an exact, free ground-truth label.

Model	Params	Zero-shot acc.
Qwen3-VL-2B-Instruct	2B	**69.3%**
Qwen2.5-VL-7B-Instruct*	7B	68.7%
Qwen2.5-VL-3B-Instruct	3B	64.9%
InternVL3-2B	2B	62.9%
Qwen2-VL-2B-Instruct	2B	62.1%
SmolVLM2-2.2B-Instruct*	2.2B	62.1%

Table 2. Zero-shot door-state accuracy of small open VLMs. *evaluated on the brighter-lighting subset.

Test set	n	Zero-shot	Fine-tuned
Door (held-out frames)	4,472	—	**97.4%**
Glass (held-out frames)	810	37.6%	**97.2%**
Door (held-out inst./swing)	7,776	84.1%	**90.2%**
Real-world doors (Diercks Hall)	100	—	**93.0%**
Real-world glass (Diercks Hall)†	100	—	**81.0%**

Table 3. Fine-tuned Qwen3-VL-2B accuracy across held-out, out-of-distribution, and real-world test sets. †The real-world glass set comprises photographs of intact glass in Diercks Hall with crack patterns applied by a diffusion model (Gemini), not photographs of physically cracked glass.

ing evaluation campaign and is the primary remaining gap before physical deployment.

4.4. Vision-Language Perception Evaluation

Model selection (zero-shot baselines). We first evaluated several small open VLMs zero-shot on synthetic door frames. No off-the-shelf model is usable as-is, and Qwen3-VL-2B is both the strongest and the smallest deployable option (Tab. 2).

Fine-tuned accuracy. After LoRA + projector co-training, the unified Qwen3-VL-2B model reaches $\mathbf{97}.4\%$ on held-out door frames ($n = 4{,}472$; open 97.4%, closed 97.4%) and $\mathbf{97.2}\%$ on held-out glass frames ($n = 810$; cracked 98.8%, intact 95.6%). The glass gain is the clearest evidence that fine-tuning, not the base model, does the work: zero-shot glass accuracy is only 37.6% (near chance), which co-training lifts to 97.2%. Table 3 summarizes performance across test sets, including the out-of-distribution and real-world transfer tests.

Generalization is the central result. An earlier single-door, single-family dataset reached high in-distribution accuracy but collapsed to $\mathbf{0}\%$ on a door instance it had never seen — it had memorized one door's appearance, not the concept of open vs. closed. This failure motivated the multi-door, multi-family, domain-randomized dataset described in Sec. 3.5. The current model retains $\mathbf{90.2}\%$ on a held-out test that mixes an unseen door instance and an unseen swing direction (up from 84.1% zero-shot), demonstrating that diverse synthetic training, not memorization, drives the result.

Real-world transfer. We tested on **100 real photographs of doors in Diercks Hall**, an existing MSOE building designed by the same architect with the same general design language as the Innovation Center. Trained only on synthetic imagery, the model reaches $\mathbf{93.0}\%$ (closed 100%, open 88.3%), confirming that the digital-twin-only training pipeline transfers to real buildings. We additionally evaluated glass on 100 Diercks Hall images in which a diffusion model (Gemini) applied crack patterns to photographs of intact glass — a semi-synthetic test, since physically cracked glass cannot be staged on demand. Here the model

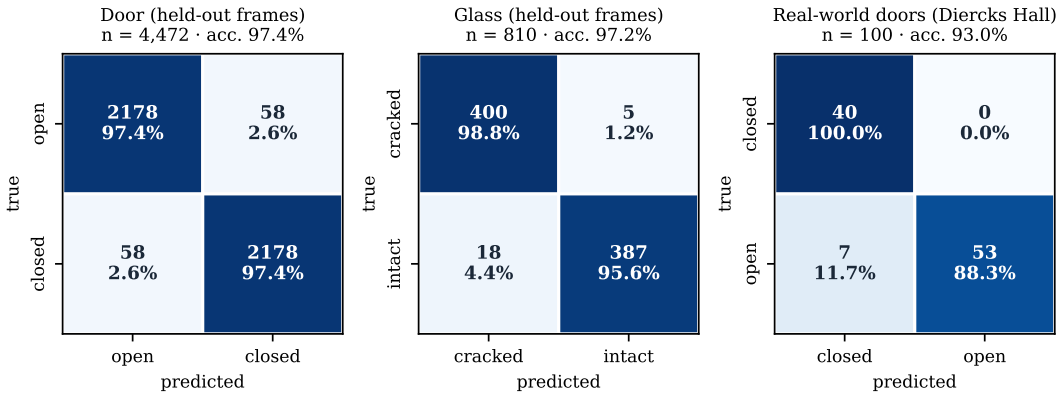


Figure 6. Confusion matrices for the fine-tuned Qwen3-VL-2B model on the held-out door, held-out glass, and real-world Diercks Hall door test sets (rows: true class; cells: count and per-class recall). The real-world split is exact; counts for the two synthetic splits are reconstructed from the reported per-class accuracies under a balanced-class assumption, which reproduces the overall accuracies of Tab. 3.

reaches **81.0**% (intact 100%, cracked 62.0%): real-glass intact frames transfer perfectly, but the diffusion-applied cracks differ enough from the twin's procedural cracks that cracked-glass recall drops, indicating the glass branch is more sensitive to the crack-appearance domain gap than the door branch.

5. Discussion

5.1. Limitations

The simulation models static building geometry without dynamic occupants (students, staff, transient furniture) that would be present during real patrol. Patrol routes are currently planar (per-floor); vertical transitions via elevators and stairwells are not yet integrated into the graph. The BIM-to-USD conversion still requires some manual intervention for certain Revit export conditions, which bounds how quickly a new building can be onboarded. Quantitative Nav2 route-execution metrics are still being tabulated. Finally, the strongest perception result is on doors; cracked-glass detection is validated on synthetic held-out frames (97.2%) and on diffusion-edited real-glass imagery (81.0%, with cracked recall falling to 62.0%), but not yet on photographs of physically cracked glass — closing that crack-appearance domain gap is future work.

5.2. Sim-to-Real Transfer

Using the same Nova Carter platform and ROS 2 command interface in simulation and deployment minimizes locomotion-dynamics mismatch, but several domain-shift sources remain: floor-surface texture/friction and minor elevation changes, sensor-noise characteristics of the physical LiDARs (not replicated in the raycaster), real-facility lighting, and — most significantly — dynamic occupants absent from the training distribution. On the perception side, the 93.0% real-world door result is strong evidence the synthetic pipeline transfers. For real deployment, robust on-boot localization is needed because the robot is initially blind to its pose; Nova Carter's LiDAR, wheel encoders, and cameras support a SLAM solution to construct and localize within a map, complementing the prior-map AMCL used in simulation.

5.3. Future Work

The most immediate extensions are quantitative Nav2 patrol evaluation in the full building, real-world cracked-glass validation, and extending the navigation graph across floors with elevators and stairwells for full-building coverage. The Kern Center recreation facility will serve as a second evaluation building and the first cross-building generalization test. Physical deployment trials on Nova Carter, supervised by the Public Safety Office, will provide the first real-world validation of the end-to-end pipeline. More broadly, the six-building digital campus is a reusable substrate for tasks well beyond patrol.

6. Conclusion

We presented a framework for automatically deriving robot training environments from Building Information Modeling data, eliminating manual scene authoring. Starting from a Revit BIM file, the pipeline converts geometry to USD, loads it into NVIDIA Isaac Sim as a physics-accurate digital twin, extracts an occupancy map and a SWAGGER navigation graph for traveling-salesman patrol planning, drives a Nova Carter robot along those routes with ROS 2 Nav2, and fine-tunes a Qwen3-VL-2B model on synthetic imagery to detect open doors and cracked glass — transferring to real-world building imagery at 93.0% accuracy. The physics pipeline is fully automated: it takes an architecture file as input and produces a physics-ready digital twin, which let us

build a **six-building digital campus** consuming over **3,500 GPU jobs and 2,000 GPU-hours** on MSOE's Rosie supercomputer. Any facility whose BIM model exists becomes a candidate for site-specific robot training before its doors ever open.

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